

Andy Cano-Avila

andycanoavila@gmail.com | www.linkedin.com/in/andy-cano-avila | Katy, TX | +1 (832) 977-2006

EDUCATION

Texas A&M University, College Station, Texas
Bachelor of Science in Mechanical Engineering

Dec 2026
3.0/4.0

WORK EXPERIENCE

Los Alamos National Laboratory Capstone - Design Engineer

College Station, TX | Aug 2025 – May 2026

- Designed a remotely hydraulic-actuated robotic hand for radioactive waste cleanup, using SolidWorks, FEA, and ISO/ANSI tolerance.
- Authored & presented various milestone reviews on the mechanical subsystem, explaining all mechanism structures and integration.
- Managed communication with our LANL/DOE customer to cement system requirements, design alterations, and safety precautions.
- Prototyped, fabricated, and tested a steel prototype capable of a 4-finger and thumb grip, fully actuating within 0.5 seconds of initiation.

Albers Aerospace - Systems Engineering Intern

Dallas, TX | May 2025 – Aug 2025

- Designed mechanical structures and contributed to design reviews (PDR, CDR, TRR, FDR) for an unmanned autonomous aircraft system.
- Implemented geometric optics and rigid-body transformations using Python-based data acquisition to support AI-driven engagements.
- Managed site testing to ensure mission success and site safety during all flight and drone-vs-drone attack scenarios.
- Rapidly developed and presented a Department of Defense design proposal within 36 hours, resulting in awarded contract funding.

Northrop Grumman - Systems Engineering Intern

McLean, VA | Jun 2024 – Aug 2024

- Led integration of Defense Systems and Strategic Deterrent divisions, streamlining cross-team processes with senior leadership.
- Analyzed RF test procedures to unify operations and deliver a heat map of improvements to executives for enhanced safety protocols.
- Consolidated 12M early-career data points into a dynamic database, optimizing recruitment and retention strategies for the company.

CLASS EXPERIENCE

Mechanical Robotic Manipulators - Control Project

College Station, TX | Mar 2026 – May 2026

- Programmed and simulated a UR5e robotic manipulator in Drake using Python, applying feedback control, gravity compensation, and external force-bias testing to study automated robot motion, eventually testing within a lab.
- Analyzed performance via generated plots and videos to assess position tracking, joint behavior, and controller effectiveness.

RELEVANT CLASSES

Mech. Robotic Manipulators | Mech. Interface Design | Solid Mech. in Design | Geometric Modeling | Dynamics & Controls | Heat Transfer

LEADERSHIP

Society of Hispanic Professional Engineers - Treasurer, Vice President

College Station, TX | May 2024 – May 2026

- Oversee a \$120,000+ chapter budget & growth initiatives for the largest SHPE chapter nationwide, serving 400+ members.
- Led legal, marketing, and finance sub-teams in planning the first SHPE run 5K in history and raising \$10.7k while engaging 362 runners.
- Drove the execution of 100+ events with an 8-officer leadership team by establishing long-term company communication pipelines.
- Planned a \$52,000 SHPE Convention trip across the country, expanding 64 members' professional opportunities and industry expertise.

SHPEathon Competition - CTO, CEO

College Station, TX | Feb 2023 / Feb 2024

- Oversaw a 6-person team in the design and marketability of an innovative, sustainable consumer product within a 48-hour window.
- Strengthened team performance by implementing conflict-resolution strategies during high-pressure design and task delegation.
- Developed a SolidWorks model with feasible integration of modern technology and pitched the product to Microsoft Reps, finishing 3rd.
- Researched viable materials, system requirements, safety protocols, and industry partners to support product viability and market fit.

ASME Go-Kart - Team Project Manager

College Station, TX | Nov 2023 – May 2024

- Directed an 11-member engineering team in modeling, simulating, fabricating, and assembling a go-kart in-house within a \$1200 budget.
- Solely responsible for welding fabrication and training the team in basic MIG welding techniques, leveraging 2 years of experience.
- Engineered and assembled the go-kart using SolidWorks and conducted FEA to validate structural integrity and weight restrictions.

NASA L'SPACE MCA - Lead Mechanical Engineer

Remote | Jan 2023 – Mar 2023

- Led mechanical subsystem design and material selection for a micro-rover, ensuring integration with science objectives and instruments.
- Designed the full CAD for the Mars rover using Autodesk Inventor, including chassis, suspension system, and scientific instruments.
- Co-authored a 146-page PDR presented to a board of NASA employees and was recognized for meeting industry-level standards.

SKILLS

Software: VScode | Docker | GitHub | SolidWorks | Drake | Virtual Machine | Microsoft Office

Skills: Data Acquisition | Python | GD&T | Tolerancing | Milling | Drilling | Welding | Soldering | Bilingual/Spanish | Project Management